

Development Tasks – Where They Fit in the Stack

Stage	What You'll Do	Tools from the Stack
Data Collection & Preprocessing	Gather sample medical, industrial, and automotive images → clean & label them	OpenCV, NumPy, Python scripts
Model Development	Train and fine-tune object/text detectors for faces, plates, documents	PyTorch, YOLOv8, EasyOCR, spaCy
Privacy Filtering Module	Implement blurring, masking, and anonymization logic	OpenCV, TensorFlow functions
Integration Layer	Build REST APIs and UI for file upload & download	Flask/FastAPI, Streamlit, React
Testing & Evaluation	Run refined test cases across all scenarios	Unit tests, test dataset scripts
Deployment	Containerize and push model to cloud or edge device	Docker, AWS Lambda, Kubernetes
Monitoring & Logging	Track privacy filter usage and compliance logs	Grafana, Prometheus, Kibana

Risks & Mitigation

Show-Stopper	Why It's a Problem	Stack-Level Mitigation
Low Detection Accuracy	Faces or plates missed due to poor lighting	Use CLAHE, data augmentation, YOLOv8 tuning
Text Extraction Errors	OCR fails on low-contrast documents	Apply adaptive thresholding + multiple OCR models
Privacy Compliance Gaps	Data sent to cloud without anonymization	Add local edge processing layer + audit logs
Processing Latency	Real-time blurring slows video feed	Use TensorRT/OpenVINO for optimized inference
Scalability & Storage	Too many large images	Use AWS S3 with compression and metadata logging
Data Security	Sensitive files stored insecurely	Encrypt with HTTPS, JWT auth, access control layer

Refined Test Cases – Connection to Stack

The refined test cases are designed to validate each functional layer of the **AI-Based Image Privacy Filter**. They ensure that the technical components — detection, redaction, compliance, and performance — work together as a unified system capable of handling real-world data securely and efficiently.

- **Face and License Detection:**

This test case validates the accuracy and robustness of the **YOLOv8 + OpenCV detection module**. It ensures that all human faces, license plates, and identifiable objects are consistently recognized across varied environments such as hospital lobbies, factory floors, and street surveillance footage. The model's confidence thresholds, bounding box accuracy, and false positive rates are monitored to confirm precise detection before the blurring process.

- **Text Redaction:**

The **OCR + NLP pipeline** is tested to verify its ability to accurately detect and classify sensitive textual content such as patient names, ID numbers, serial codes, or addresses. This test ensures that the system not only identifies visible text but also distinguishes between sensitive and non-sensitive content, so only the necessary parts are blurred while retaining the rest of the information for context.

- **Compliance and Data Handling:**

This test case validates the **audit, encryption, and data governance layers** of the stack. It ensures that image data is processed securely, anonymized locally before cloud upload, and logged with time-stamped audit trails. The goal is to meet privacy compliance standards like **HIPAA, GDPR, and DPDP**, proving that the system maintains confidentiality even when data is transmitted across networks or shared with external partners (e.g., hospitals and insurance firms).

- **System Performance:**

The final validation focuses on overall **system latency, resource usage, and scalability**. It tests how efficiently the model performs under high-load conditions, such as live CCTV feeds or large-scale hospital datasets. The use of **TensorRT and OpenVINO** ensures low-latency inference, and the system is stress-tested to confirm stable performance without compromising detection or privacy accuracy.